

Classification algorithm

logistic regression:

x	y
1	0
2	0
3	1
4	1

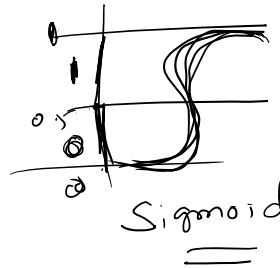
yes
no

Binary Classification

$$y = \beta_1 x + \beta_0$$

$$y = \text{model}$$

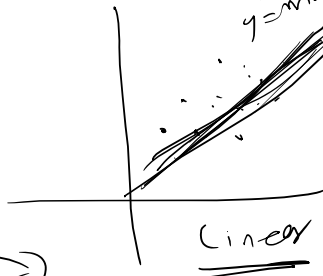
x ₁	y
20	21
18	23
19	24
17	20
...	...



1.5 $\rightarrow$ 2

$$\frac{1}{2} = 0.5$$

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First stage $y = \beta_1 x + \beta_0$

$$= \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-\beta_1 x + \beta_0}}$$

x	y
1	0
2	0
3	1
4	1

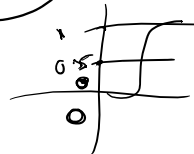
1. 0.26

0. 0.5

1. 0.74

2. 0.74

$$y = 2 + 1x$$



$$= \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\beta_1 x + \beta_0)}} = 0.26$$

LR $\rightarrow$ output $= \frac{1}{1 + e^{-z}}$ Sigmoid (logit)

$\rightarrow$ Logistic Regression

log-odds

h $\rightarrow$ $p + 2 = 1$ $h = \frac{1}{2}$

log-odds

(h) (f)

$$p + z = 1$$

$$h = \frac{1}{2}$$

$$1 - \frac{1}{2} = \frac{1}{2}$$

$$\log\left(\frac{p}{1-p}\right) = \beta_1 x + \beta_0$$

$$\frac{p}{1-p} = e^{\beta_1 x + \beta_0}$$

$$p = \left(e^{\beta_1 x + \beta_0} \right) (1-p)$$

$$p = \frac{e^{\beta_1 x + \beta_0}}{1 + e^{\beta_1 x + \beta_0}} = p_0 e^{\beta_1 x + \beta_0}$$

$$p + p_0 e^{\beta_1 x + \beta_0} = e^{\beta_1 x + \beta_0}$$

$$p_0 (1 + e^{\beta_1 x + \beta_0}) = e^{\beta_1 x + \beta_0}$$

$$p = \frac{e^{\beta_1 x + \beta_0}}{1 + e^{\beta_1 x + \beta_0}} = \frac{e^{\beta_1 x + \beta_0}}{1 + e^{\beta_1 x + \beta_0}}$$

$$\frac{1}{n} = \frac{1}{n}$$

$$= \frac{1}{\frac{1}{e^{\beta_1 x + \beta_0}} + 1} = \frac{1}{e^{-\beta_1 x + \beta_0} + 1}$$

$$y = \omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + \omega_4 x_4 + \omega_5 x_5 + \omega_6 x_6 + \omega_7 x_7 + \omega_8 x_8 + \omega_0$$

$$= 0.06 x_1 + 0.03 x_2 + 0.01 x_3 + 0.003 x_4 + 0.001 x_5 + 0.10 x_6 + 0.62 x_7 + 0.03 x_8 + 9.0$$

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